B25ET1005 OOP, Practical-1

FAQ's

1) What is Pascal's triangle?
→ A triangular array of numbers where each entry is the sum of the two numbers directly above it.

2) Why are nested loops suitable for this program?
→ They directly mirror the 2D layout of Pascal's Triangle by using an outer loop for rows and an inner loop for columns.

3) Relatio What is the relationship between Pascal's triangle & binomial coefficients?
→ The k-th entry in a row n of Pascal's Triangle corresponds to the binomial coefficient $\binom{n}{k}$, representing the coefficients of $(a+b)^n$.

4) What are different features provided by OOP languages as opposed to procedural languages?
→ OOP provides encapsulation, inheritance, polymorphism & abstraction to organize code around objects rather than linear step-by-step procedures.

5) What do you mean by runtime initialization of variables?
→ Assigning a value to a variable while the program is actively executing rather than hard coding it at compile time.

6) What is difference between function definition, call & declaration?

→ A declaration tells the compiler a function exists, definition provides the actual code implementation, & a call invokes & executes that function during runtime.